

WASP-10b

R-2281

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_110922 · created 2026-08-17T00:10:47+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455826.9057 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.21 +/- 0.061
Transit depth (Rp/Rs)^2	4.43 +/- 2.59 [%]
Orbital Inclination (inc)	89.72 +/- 1.83
Airmass coefficient 1 (a1)	0.96 +/- 0.12
Airmass coefficient 2 (a2)	0.07 +/- 0.065
Scatter in the residuals of the lightcurve fit is	13.49 %
Best Comparison Star	#2 - [289, 169]
Optimal Aperture	2.86
Optimal Annulus	9.8
Transit Duration (day)	0.095 +/- 0.0085

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.21 ± 0.061 vs 0.1592 (deviation 0.64σ)
Duration fit vs published	0.095 d vs 0.0946 d (deviation 0.04σ)
Mid-time O–C	2.4 min
Depth SNR	2.6
Residual scatter	13.49 %

Light curve

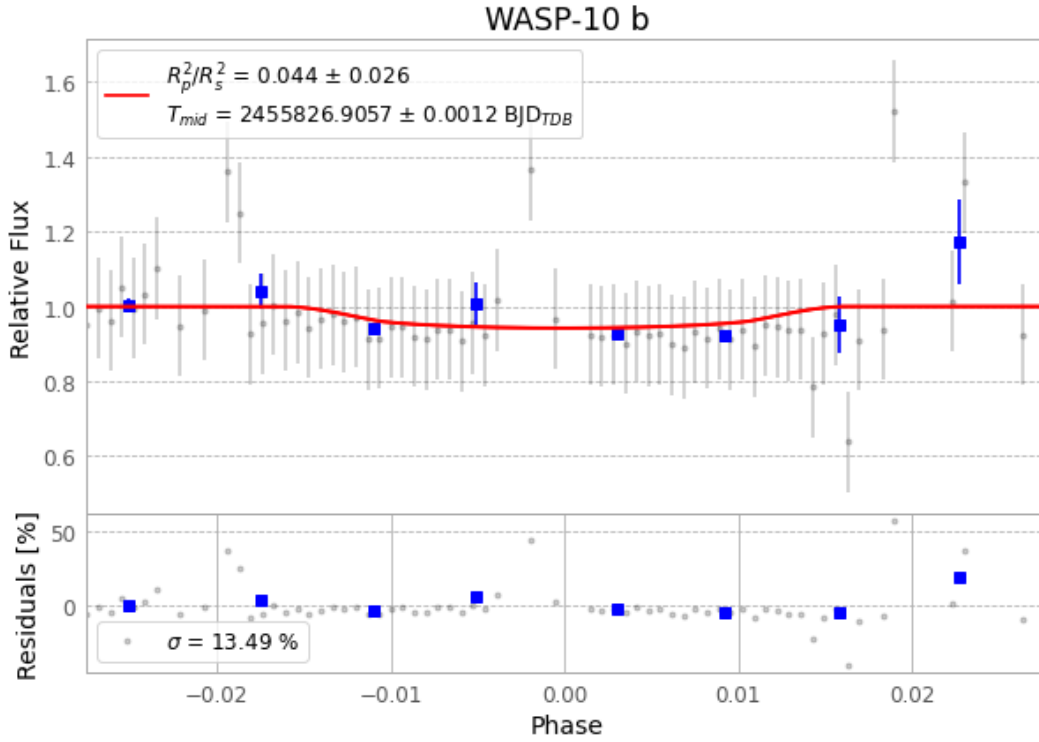


Figure 1 — FinalLightCurve_WASP-10 b_2011-09-22.png (SHA-256 3364ee2ce4dda078...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [348, 213], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 0586bfc8ef0fd398...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

83 FITS frames (54 MB), WASP-10110922073312.FITS → WASP-10110922113912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-110922.log (6642662d4d12...), FinalLightCurve_WASP-10 b_2011-09-22.png (3364ee2ce4dd...), FinalLightCurve_WASP-10 b_2011-09-22.csv (c61fde7067ae...)

Compute resources

Wall time 307.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 00:10Z.